Sahaj Bhadja

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Summary

Cloud-focused software engineer with 2+ years building backend services, REST integrations, asynchronous workflows, and production data systems. Experienced in Python, Node.js, AWS, SQL, CI/CD, automated testing, monitoring, incident troubleshooting, and reliable cloud operations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and operational dashboards for 500+ users, tuning indexed queries and secure access paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented asynchronous processing for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving transaction recovery, observability, maintainability, and production incident diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, implementing secure file ingestion, authentication, job-state tracking, and automated grading workflows while resolving production issues for 150+ users.
- Created concurrent processing for 1,000+ files with automated validation, AWS RDS persistence, structured logging, and failed-job recovery, reducing turnaround 60% while maintaining reliable behavior under production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and event-driven ETL workflows using AWS Glue, S3, and REST APIs, improving response times 30% and data-processing efficiency 68% across production cloud applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating root-cause analysis, release validation, and reliable cloud production deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight services with JWT authentication and role-based access controls, synchronizing 2 million+ Azure SQL records through secure APIs and real-time WebSockets.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and operational visibility across environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs and responsive workflows for reservations, status updates, and payments across 50,000+ records, translating operational requirements into maintainable backend features with validation and secure Firebase access.
- Implemented Firebase persistence, webhooks, and payment synchronization with structured validation and logging, reducing manual payment handling 70% while improving failed-transaction visibility, integration reliability, and responsive production support.

Skills

Languages & Backend: Python, Java, JavaScript, TypeScript, SQL, Node.js, FastAPI, Flask, REST APIs, microservices, WebSockets, asynchronous processing
Cloud & Integration: AWS, AWS RDS, AWS Glue, Amazon S3, Azure, Blob Storage, event-driven workflows, webhooks, API integration, distributed workflows, transaction recovery
Data: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, relational databases, joins, data modeling, filtering, pagination, query optimization
DevOps & Operations: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, automated testing, debugging, logging, monitoring, observability, incident troubleshooting, root-cause analysis, production support, Agile

Projects

FinNexus | Live | *TypeScript, REST APIs, WebSockets, MongoDB, Kubernetes, CI/CD*
- Built cloud-native financial workflows supporting 50,000+ users and one million transactions, integrating REST APIs, asynchronous synchronization, MongoDB persistence, monitoring, validation, and automated Kubernetes deployment through production-ready CI/CD pipelines.

Chat2SQL | Code | *Java, TypeScript, PostgreSQL, MySQL, MongoDB, Docker*
- Built a Dockerized backend tool supporting natural-language queries across PostgreSQL, MySQL, and MongoDB, implementing REST APIs, input validation, relational joins, and reproducible Maven builds for reliable deployment.

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built monitored operational workflows for appointments, records, and notifications using React, REST APIs, Appwrite, Twilio, and Sentry observability, supporting 100+ daily interactions while reducing administrative workload 30%.